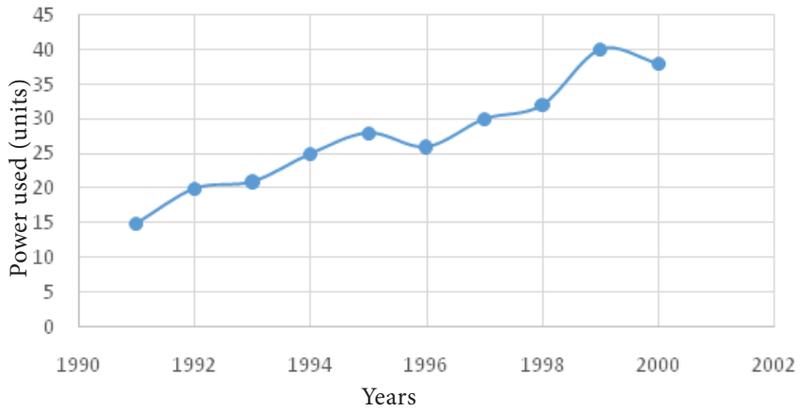
Solution:



7.3.2 Semi-Average Method

In this method, the series is divided into two equal parts and the average of each part is plotted at the mid-point of their time duration.

(i) In case the series consists of an even number of years, the series is divisible into two halves. Find the average of the two parts of the series and place these values in the mid-year of each of the respective durations.

(ii) In case the series consists of odd number of years, it is not possible to divide the series into two equal halves. The middle year will be omitted. After dividing the data into two parts, find the arithmetic mean of each part. Thus we get semi-averages.

(iii) The trend values for other years can be computed by successive addition or subtraction for each year ahead or behind any year.

Merits

- This method is very simple and easy to understand
- It does not require many calculations.

NOTE

In semi-average method if the difference between the semi-averages is negative then the trend values will be in decreasing order.

Demerits

- This method is used only when the trend is linear.
- It is used for calculation of averages and they are affected by extreme values.

Example 7.2

Calculate the trend values using semi-averages methods for the income from the forest department. Find the yearly increase.

Year	2008	2009	2010	2011	2012	2013
Income (in crores)	46.17	51.65	63.81	70.99	84.91	91.64

Source: The Principal Chief conservator of forests, Chennai-15. (pg. 231)

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